

COMICCRAFT – TEST CASES

1. Objective

The objective of the test cases is to verify each major function of the ComicCraft application and ensure that the actual results match the expected results.

2. Test Case Format

Each test case contains:

- Test Case ID
- Test Scenario
- Test Input
- Expected Result
- Actual Result
- Status

3. Test Cases

TC01 – Application Launch

Test Scenario: Verify that the ComicCraft application opens successfully.

Test Input: Start the FastAPI application and open it in a web browser.

Expected Result: The ComicCraft home page should be displayed.

Actual Result: Application opens and displays the home page.

Status: Pass

TC02 – Story Prompt Input

Test Scenario: Verify that the user can enter a story prompt.

Test Input: Enter a valid comic story idea.

Expected Result: The story prompt should be accepted.

Actual Result: The prompt is accepted.

Status: Pass

TC03 – Empty Prompt Validation

Test Scenario: Verify that an empty prompt is rejected.

Test Input: Submit the form without entering a story prompt.

Expected Result: The application should display an appropriate validation message.

Actual Result: Empty prompt is rejected with an error message.

Status: Pass

TC04 – Story Generation

Test Scenario: Verify that the AI generates a story from a valid prompt.

Test Input: Enter a valid story idea and select the generate option.

Expected Result: The application should generate a story outline and comic content.

Actual Result: AI-generated story content is received.

Status: Pass

TC05 – Comic Panel Generation

Test Scenario: Verify that the generated story is divided into comic panels.

Test Input: Use a valid story prompt.

Expected Result: Multiple structured comic panels should be generated.

Actual Result: Comic panels are generated with required information.

Status: Pass

TC06 – Image Prompt Generation

Test Scenario: Verify that an image prompt is created for each comic panel.

Test Input: Generate a comic story.

Expected Result: Each panel should contain an image prompt.

Actual Result: Image prompts are generated for the panels.

Status: Pass

TC07 – Image Generation

Test Scenario: Verify that the image generation module processes panel image prompts.

Test Input: Generate a comic containing multiple panels.

Expected Result: Corresponding panel images should be generated or handled using the available image-generation workflow.

Actual Result: Panel images are processed and associated with the corresponding panels.

Status: Pass

TC08 – Comic Layout

Test Scenario: Verify that text and images are combined into the comic layout.

Test Input: Generated story panels and panel images.

Expected Result: A structured comic layout should be created.

Actual Result: Comic layout is created successfully.

Status: Pass

TC09 – PDF Generation

Test Scenario: Verify that the comic can be converted into a PDF.

Test Input: Completed comic layout.

Expected Result: A PDF file should be generated and saved.

Actual Result: PDF file is generated and stored.

Status: Pass

TC10 – PDF Access

Test Scenario: Verify that the generated PDF can be accessed by the user.

Test Input: Select the PDF view/download option.

Expected Result: The generated PDF should be available to the user.

Actual Result: Generated PDF is available.

Status: Pass

TC11 – Invalid Request Handling

Test Scenario: Verify that invalid requests are handled properly.

Test Input: Send invalid or incomplete request data.

Expected Result: The application should return an appropriate error response.

Actual Result: Invalid request is handled with an error response.

Status: Pass

TC12 – Missing File Handling

Test Scenario: Verify that the application handles missing image or PDF files.

Test Input: Request a file that is unavailable.

Expected Result: The application should display an appropriate error instead of crashing.

Actual Result: Missing file is handled with an error response.

Status: Pass

4. Test Case Summary

Test Case	Description	Status
TC01	Application Launch	Pass
TC02	Story Prompt Input	Pass
TC03	Empty Prompt Validation	Pass
TC04	Story Generation	Pass
TC05	Comic Panel Generation	Pass
TC06	Image Prompt Generation	Pass

TC07	Image Generation	Pass
TC08	Comic Layout	Pass
TC09	PDF Generation	Pass
TC10	PDF Access	Pass
TC11	Invalid Request Handling	Pass
TC12	Missing File Handling	Pass

5. Conclusion

The test cases cover the major features and components of the ComicCraft application.

These test cases help verify the complete workflow from entering a story prompt to generating the comic and exporting it as a PDF.